Holistic P.E.N. Way™ Neurodevelopmental, Sensory & Regulation Wellness Assessment

Practitioner Framework for Autism, Neurodivergence, Sensory Processing Differences, Behavioral Dysregulation & Developmental Concerns

Purpose:
This assessment is designed to help the holistic wellness practitioner identify areas of physiological, nutritional, sensory, emotional, environmental, developmental, and lifestyle stress that may be affecting a child's overall function.

It is **not intended to diagnose autism, genetic disorders, parasites, neurologic disease, allergies, infections, endocrine disorders, or psychiatric conditions.**

NLS/biofeedback findings are recorded as **wellness observations or patterns for further investigation**, not as medical diagnoses.

I. FOUNDATIONAL P.E.N. WAY™ PRINCIPLE

The goal is not:

“What is wrong with this child?”

The goal is:

“What factors may be increasing the load on this child's developing nervous system, and what supports may help the child function, regulate, communicate, nourish, sleep, learn, and feel safe?”

Autism is a heterogeneous neurodevelopmental condition. There is no scientifically established single parasite, toxin, vitamin deficiency, traumatic experience, infection, or metabolic abnormality that explains all autism.

At the same time, autistic and otherwise neurodivergent children can have significant **co-occurring medical, nutritional, neurologic, sensory, gastrointestinal, sleep, genetic, emotional, and environmental problems** that deserve individual investigation.

The American Academy of Pediatrics specifically recommends evaluation for co-occurring medical and behavioral conditions including sleep problems, feeding problems, GI symptoms, seizures, ADHD, anxiety, and other conditions. Its autism clinical report was reaffirmed in October 2025.

II. IMMEDIATE RED-FLAG SCREEN

Before building an herbal, nutritional, biofeedback, sensory, or somatic wellness plan, determine whether medical evaluation is needed.

REFER PROMPTLY FOR:

☐ New seizures or suspected seizures
☐ Repetitive staring with loss of awareness
☐ Loss of consciousness
☐ Sudden loss of motor ability
☐ Significant or unusual developmental regression
☐ New weakness or one-sided neurologic symptoms
☐ Severe headaches with vomiting
☐ Abnormal gait developing suddenly
☐ Persistent unexplained fever
☐ Significant weight loss or failure to grow
☐ Persistent vomiting or diarrhea
☐ Blood in stool
☐ Severe abdominal pain
☐ Severe dehydration
☐ Suspected poisoning or significant environmental exposure
☐ New vision loss
☐ Severe behavioral change accompanied by physical illness
☐ Suicidal behavior or serious self-injury

Routine EEG testing is not recommended simply because a child has autism, but neurologic symptoms, suspected seizures, or unusual regression are reasons for appropriate neurologic evaluation.

III. DEVELOPMENTAL TIMELINE

Create a timeline rather than beginning with supplements.

Pregnancy

Record:

☐ Maternal illnesses
☐ Significant infections
☐ Fever during pregnancy
☐ Nutritional difficulties
☐ Severe vomiting/hyperemesis
☐ Prescription medication exposures
☐ Alcohol exposure

☐ Nicotine exposure
☐ Opioid exposure
☐ Stimulant/methamphetamine exposure
☐ Other substance exposure
☐ Diabetes
☐ Thyroid problems
☐ Severe stress or traumatic events
☐ Premature labor
☐ Known infections such as toxoplasmosis

Prenatal exposures can influence neurodevelopment, but they should be documented as **risk factors**, not assumed to be the explanation for autism.

Birth

Record:

☐ Gestational age
☐ Birth weight
☐ C-section/vaginal delivery
☐ Prolonged labor
☐ Oxygen complications
☐ NICU admission
☐ Neonatal withdrawal
☐ Feeding difficulties
☐ Jaundice
☐ Infection
☐ Seizures
☐ Birth trauma

Development

Document approximately when the child:

☐ Sat
☐ Crawled
☐ Walked
☐ Spoke first words
☐ Combined words
☐ Pointed
☐ Responded to name
☐ Used pretend play
☐ Toilet trained

Regression

Ask specifically:

“Did this child ever have a skill and then lose it?”

Document:

- Exact skill lost
- Approximate age
- Whether loss was sudden or gradual
- Illness around that period
- Medication changes
- Seizure-like episodes
- Traumatic events
- Major environmental changes

Developmental regression deserves medical evaluation even though regression can occur in autism without another identifiable neurologic disease.

IV. GENETIC & CHROMOSOMAL ROUTE

There is no human “B chromosome.”

Humans normally have **22 pairs of autosomes plus the sex chromosomes X and Y**.

Keep two pathways separate:

A. Genetic/chromosomal pathway

and

B. B-vitamin/folate/methylation pathway.

They can interact biologically, but they are not the same thing.

FAMILY HISTORY

Ask about:

☐ Autism
☐ ADHD
☐ Intellectual disability
☐ Developmental delay

☐ Speech delay
☐ Learning disability
☐ Epilepsy
☐ Congenital abnormalities
☐ Fragile X syndrome
☐ Rett syndrome
☐ Tuberous sclerosis
☐ Known chromosome deletion/duplication
☐ Unexplained infant/child deaths
☐ Significant psychiatric or neurologic disorders

Clinical genetics pathway

Depending upon the child's presentation, conventional evaluation may include:

Chromosomal microarray (CMA)
Detects certain deletions and duplications of chromosome material.

FMR1/Fragile X testing

Targeted genetic testing
When a recognizable syndrome is suspected.

Exome/genome sequencing
Increasingly used when the clinical picture suggests a genetic cause and earlier testing is unrevealing.

Current primary-care guidance recommends chromosomal microarray or exome sequencing for children with autism or global developmental delay, with Fragile X testing particularly considered in males.

NLS chromosome observations

An NLS chromosome/genetic panel may be recorded as:

“NLS biofeedback observation suggesting an area for further wellness investigation.”

Do NOT document:

“Chromosome abnormality confirmed.”

Instead document:

> NLS screening produced an observation associated with the chromosome/genetic panel. Family was advised that biofeedback cannot establish a chromosome deletion, duplication, mutation, or genetic syndrome and that clinical genetic testing is required if concern remains.

V. B-VITAMIN / FOLATE / ONE-CARBON METABOLISM ROUTE

Investigate separately from chromosomes.

Consider:

☐ Restricted diet
☐ Food selectivity
☐ Low animal-protein intake
☐ Poor appetite
☐ Malabsorption concerns
☐ Chronic diarrhea
☐ Constipation
☐ Fatigue
☐ Pallor
☐ Tongue/oral changes
☐ Neurologic symptoms

Possible conventional laboratory questions may include:

- CBC
- Ferritin/iron studies
- Vitamin B12
- Folate
- Homocysteine
- Vitamin D
- Other nutrients based upon diet and clinical findings

Do not interpret a common genetic variant such as an MTHFR polymorphism by itself as “the cause of autism.”

Instead ask:

Does this particular child have evidence of impaired nutritional sufficiency or metabolism that deserves correction?

VI. NEUROLOGIC & CENTRAL NERVOUS SYSTEM ROUTE

Review:

☐ Head injury
☐ Concussion
☐ Falls
☐ Birth injury
☐ Seizures
☐ Staring spells
☐ Abnormal movements
☐ Tremors
☐ Motor delay
☐ Coordination problems
☐ Headaches
☐ Regression
☐ Sleep disturbance
☐ Muscle tone
☐ Balance

NLS panels you may review as observational information

- Cerebrum
- Cerebellum
- Limbic-associated panels
- Brainstem-associated panels
- Hypothalamus
- Pituitary
- Autonomic nervous system
- Peripheral nervous system

However:

An NLS pattern involving the hypothalamus or pituitary **does not diagnose hypothalamic or pituitary injury**.

Persistent clinical signs require conventional assessment.

VII. SENSORY PROCESSING ROUTE

Separate sensory overload from intentional misbehavior.

Evaluate the child's responses to:

Auditory

☐ Vacuum cleaners
☐ Toilets/hand dryers
☐ Crowds

☐ Sudden noises
☐ Multiple people talking
☐ High-pitched sounds

Visual

☐ Bright fluorescent lights
☐ Flashing lights
☐ Busy rooms
☐ Certain colors/patterns

Touch

☐ Clothing tags
☐ Socks/shoes
☐ Hair brushing
☐ Bathing
☐ Hugging
☐ Light touch
☐ Grooming

Oral/feeding

☐ Food texture
☐ Food temperature
☐ Color preference
☐ Chewing
☐ Strong flavors
☐ Oral seeking

Vestibular/proprioceptive

☐ Spinning
☐ Jumping
☐ Crashing
☐ Rocking
☐ Swinging
☐ Constant movement
☐ Avoidance of movement

Interoception

Does the child recognize:

☐ Hunger
☐ Thirst
☐ Full bladder

☐ Need for bowel movement
☐ Hot/cold
☐ Fatigue
☐ Pain
☐ Rising emotional activation

A 2025 systematic review found that properly delivered Ayres Sensory Integration supported individualized occupational goals and participation in autistic children, although it was not shown to broadly eliminate behaviors such as irritability.

Consider referral to pediatric occupational therapy when sensory difficulties substantially interfere with eating, dressing, toileting, learning, sleep, grooming, or daily participation.

VIII. AUTONOMIC / VAGUS / REGULATION ROUTE

Do not frame every episode as behavioral defiance.

Ask:

What happened immediately BEFORE the episode?

☐ Loud sound
☐ Unexpected transition
☐ Hunger
☐ Constipation
☐ Pain
☐ Lack of sleep
☐ Too many instructions
☐ Social demand
☐ Clothing discomfort
☐ Smell
☐ Bright lights
☐ Fear
☐ Conflict
☐ Separation
☐ Trauma reminder
☐ Communication failure

Observe:

☐ Rapid breathing
☐ Breath holding
☐ Sweating
☐ Flushing
☐ Pale appearance

☐ Trembling
☐ Clenching
☐ Pacing
☐ Running
☐ Freezing
☐ Hiding
☐ Shutting down
☐ Sudden sleepiness
☐ GI urgency
☐ Nausea

Regulation supports

Possible low-risk supportive strategies include:

- predictable routine
- preparation for transitions
- sensory breaks
- movement
- outdoor activity
- rhythmic movement
- quiet spaces
- reduced sensory load
- visual schedules
- co-regulation
- child-selected music
- singing or humming when enjoyed
- slow breathing when tolerated
- proprioceptive activities recommended by OT

Use the term:

autonomic regulation support

rather than promising to “reset the vagus nerve.”

IX. TRAUMA & SOMATIC ROUTE

Trauma can exist alongside autism, ADHD, sensory differences, intellectual disability, prenatal exposure, attachment disruption, and other developmental conditions.

Do not automatically interpret autism as trauma.

Do not automatically interpret trauma behaviors as autism.

Consider:

☐ Prenatal adversity
☐ NICU/medical trauma
☐ Hospitalization
☐ Foster care
☐ Adoption transition
☐ Loss of caregiver
☐ Domestic violence exposure
☐ Abuse/neglect
☐ Housing instability
☐ Food insecurity
☐ Bullying
☐ School trauma
☐ Restraint/seclusion history
☐ Death or major loss
☐ Other frightening events

Ask:

“Was there a noticeable change in behavior, sleep, eating, toileting, social engagement, or sensory tolerance after a specific event?”

Recent reviews emphasize that trauma symptoms may be difficult to recognize in autistic children because trauma symptoms can overlap with communication and sensory differences.

Trauma processing itself should remain within the scope of appropriately trained/licensed professionals when PTSD, abuse, serious dissociation, suicidality, or complex psychiatric symptoms are suspected.

X. GUT–BRAIN / DIGESTIVE ROUTE

Always screen for discomfort before labeling a behavior problem.

Assess:

☐ Constipation
☐ Diarrhea
☐ Reflux
☐ Abdominal pain
☐ Gas
☐ Bloating
☐ Stool withholding
☐ Painful bowel movements
☐ Poor appetite
☐ Excessive appetite

☐ Food selectivity
☐ Texture avoidance
☐ Chewing/swallowing concerns

A child who cannot reliably communicate pain may express constipation, dental pain, reflux, infection, headache, or other discomfort through behavioral changes.

XI. ALLERGY / FOOD / ENVIRONMENTAL ROUTE

Document actual reaction patterns.

Ask:

What happens?

Examples:

- hives
- throat symptoms
- wheezing
- vomiting
- diarrhea
- eczema
- swelling
- headache
- behavioral changes temporally associated with exposure

How soon after exposure?

Does the reaction occur every time?

NLS or electrodermal findings should not be considered confirmation of a food allergy.

Suspected IgE-mediated food allergy should be evaluated through appropriate allergy assessment.

Do not eliminate many foods solely because an NLS panel flags them. Highly restrictive diets may worsen nutritional deficiencies.

NCCIH notes that special diets have mixed evidence in autism and recommends monitoring nutritional adequacy carefully.

XII. PARASITIC & INFECTIOUS ROUTE

This pathway should be **symptom- and exposure-driven**.

Ask about:

☐ Travel
☐ Camping
☐ Well water
☐ Untreated water
☐ Soil exposure
☐ Pica
☐ Barefoot soil exposure
☐ Pets/animal exposure
☐ Cat litter
☐ Raw/undercooked meat
☐ Unwashed produce
☐ Household members with parasites
☐ Chronic GI symptoms
☐ Anal itching
☐ Anemia
☐ Eosinophilia
☐ Unexplained weight loss
☐ Persistent diarrhea

An NLS indication of:

- hookworm
- pinworm
- roundworm
- Giardia
- Toxoplasma
- other organism

should generate a **clinical question**, not an antiparasitic diagnosis.

Example:

> NLS observation produced a pattern corresponding with a parasitic/infectious panel. Exposure and symptom history were reviewed. Clinical confirmation through an appropriate healthcare professional was recommended before antimicrobial or antiparasitic treatment.

XIII. TOXOPLASMA GONDII ROUTE

Toxoplasma gondii deserves special attention because congenital infection is a real neurologic condition.

CDC recognizes congenital toxoplasmosis as capable of affecting the brain and eyes and producing developmental delay, seizures, abnormal head size, hearing problems, and other neurologic manifestations.

Potential exposure includes:

- maternal infection during pregnancy
- undercooked contaminated meat
- contaminated soil
- unwashed produce
- contaminated water
- exposure to infected cat feces

CDC notes that most healthy people who acquire toxoplasmosis have either no symptoms or mild illness, while congenital infection is a different clinical situation.

If NLS indicates Toxoplasma

Review:

☐ Pregnancy history
☐ Maternal infection history
☐ Vision problems
☐ Seizures
☐ Abnormal neurologic development
☐ Head-size abnormalities
☐ Known prenatal findings

Then refer for appropriate conventional testing if clinically warranted.

Do not tell the family:

“Toxoplasma is causing the autism.”

Document instead:

“An infectious pathway merits additional investigation.”

XIV. ENDOCRINE / HORMONAL ROUTE

NLS observational review may include:

- hypothalamus
- pituitary
- thyroid
- adrenal-associated panels
- pancreatic/metabolic panels

Clinical questions:

☐ Growth abnormality
☐ Excessive thirst
☐ Excessive urination
☐ Significant weight change
☐ Temperature intolerance
☐ Abnormal puberty
☐ Persistent severe fatigue
☐ Constipation
☐ Tremor/tachycardia
☐ Sleep disruption

Abnormal clinical findings should trigger appropriate laboratory/medical evaluation rather than an endocrine diagnosis from biofeedback alone.

XV. NLS BIOFEEDBACK REVIEW WORKSHEET

For each observation, record four things:

NLS observation	Child's real-world symptoms/history	Question generated	Confirmation/referral
Chromosome/ genetic panel	Developmental/family history	Could a genetic condition be present?	Genetics/CMA/FMR1/ exome as indicated
B-vitamin pathway	Restricted diet/fatigue	Nutrient insufficiency?	Dietary assessment/labs
Brain/CNS panel	Regression/staring/ head injury	Neurologic condition?	Pediatrician/neurology
Pituitary/ hypothalamus	Growth/endocrine symptoms	Endocrine dysfunction?	Clinical endocrine workup
Parasite panel	GI/exposure symptoms	Is infection plausible?	Organism-specific testing

NLS observation	Child's real-world symptoms/history	Question generated	Confirmation/referral
Toxoplasma	Prenatal/exposure/ neurologic history	Congenital/acquired infection concern?	Appropriate serology/ medical evaluation
Allergy panel	Reproducible reaction	True allergy?	Allergist/testing
GI panel	Constipation/reflux/pain	GI disorder?	Pediatric/GI evaluation
Thyroid panel	Growth/fatigue/ temperature	Thyroid dysfunction?	TSH/free T4 etc.
Autonomic panel	Overload/shutdown/ fight-flight	Regulation difficulty?	Clinical/sensory assessment

XVI. HERBAL SUPPORT ROUTE

The herbal goal is **not to treat autism**.

The herbal question is:

“Does this individual child have a specific symptom for which an age-appropriate botanical might safely provide supportive comfort?”

Potential symptom categories:

Relaxation / settling

Gentle botanicals sometimes considered in pediatric herbal practice include:

- chamomile
- lemon balm

These still require consideration of:

- age
- body size
- allergies
- medication interactions
- seizure history
- medical conditions
- preparation strength

Digestive comfort

Herbal selection should depend upon the actual complaint:

- constipation is different from diarrhea
- reflux is different from intestinal cramping
- food allergy is different from indigestion
- infection is different from functional GI distress

Antiparasitic herbs

Do not automatically begin strong antiparasitic herbs in a child because an NLS panel indicates a parasite.

Confirm whether the organism is actually present when possible.

Aggressive antimicrobials can cause adverse effects, interactions, liver injury, GI irritation, or inappropriate delays in treatment.

XVII. AROMATHERAPY

Sensory preference takes priority.

For some neurodivergent children, fragrance is soothing.

For others it is an intense sensory stressor.

Ask:

“Does the child actually enjoy this smell?”

Avoid:

- forced inhalation
- essential oil ingestion
- undiluted application
- application near eyes/nose
- placing oils where the child can ingest them

Stop immediately with:

- wheezing
- coughing
- headache
- nausea

- skin reaction
- increased agitation

XVIII. PATCHES

Kind Patch, LifeWave, PatchAid or similar patches should be categorized as:

optional complementary wellness tools

rather than treatments for:

- autism
- genetic abnormalities
- parasites
- brain injury
- chromosome abnormalities
- neurologic disease

Before use:

☐ Check ingredients
☐ Check medication/supplement duplication
☐ Assess adhesive sensitivity
☐ Patch-test when appropriate
☐ Monitor behavioral and skin response
☐ Remove immediately with significant reaction

A vitamin patch should not replace evaluation and treatment of a clinically significant nutrient deficiency.

XIX. NUTRITION FIRST

Before an elaborate supplement protocol, answer these questions:

Is the child:

☐ Eating enough calories?
☐ Eating enough protein?
☐ Drinking enough water?
☐ Constipated?
☐ Sleeping?
☐ Growing appropriately?
☐ Eating fewer than approximately 10–20 foods?

☐ Avoiding entire food groups?
☐ Showing signs of feeding dysfunction?

NCCIH emphasizes that children using restrictive/special diets should have nutritional status monitored because food restriction can create nutritional problems.

XX. SLEEP ROUTE

Document:

☐ Bedtime
☐ Sleep onset
☐ Night waking
☐ Snoring
☐ Mouth breathing
☐ Restless sleep
☐ Night terrors
☐ Early waking
☐ Daytime sleepiness

Address routine, light exposure, sensory environment, physical activity, pain, constipation, and sleep-disordered breathing before assuming the problem is purely behavioral.

Melatonin has evidence for some sleep difficulties in autistic children, but NCCIH recommends clinician involvement because pediatric dosing, side effects, product quality, and long-term safety require consideration.

XXI. BEHAVIOR = COMMUNICATION WORKSHEET

Whenever there is a significant episode, document:

BEFORE

What happened immediately beforehand?

BODY

Was the child:

☐ Hungry
☐ Tired
☐ Constipated

☐ Sick
☐ In pain
☐ Overheated
☐ Overstimulated

SENSORY

Was there:

☐ Noise
☐ Smell
☐ Touch
☐ Bright light
☐ Crowd
☐ Clothing discomfort

EMOTIONAL

Was there:

☐ Fear
☐ Separation
☐ Unexpected change
☐ Trauma reminder
☐ Frustration

COMMUNICATION

Was the child unable to communicate:

☐ Stop
☐ No
☐ Help
☐ Pain
☐ Toilet
☐ Hungry
☐ Break
☐ Too loud
☐ Too much

AFTER

What actually helped the child's nervous system recover?

This often reveals more useful information than labeling an event a “meltdown,” “tantrum,” or “bad behavior.”

XXII. P.E.N. WAY™ INDIVIDUALIZED PLAN

P — PERSPECTIVE

Spirit • Identity • Environment • Meaning • Safety

Goals:

- honor the child's individual identity
- recognize strengths
- create safety and predictability
- support the family
- reduce shame
- distinguish inability from unwillingness
- provide developmentally appropriate spiritual support when desired by the family

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E — ESSENCE

Soul • Emotional Regulation • Sensory Processing • Autonomic Regulation • Trauma-Informed Support

Possible priorities:

- sensory profile
- predictable routines
- transition support
- communication tools
- OT referral
- movement
- co-regulation
- sleep regulation
- trauma-informed care
- somatic regulation
- individualized autonomic support

N — NUTRITION

Body • Nourishment • Gut • Biochemical Sufficiency • Physical Comfort

Priorities:

1. hydration
2. adequate calories

3. adequate protein
4. regular bowel movements
5. adequate sleep
6. feeding safety
7. nutrient sufficiency
8. documented deficiency correction
9. GI symptom management
10. targeted herbs/supplements only when appropriate

XXIII. FOLLOW-UP MEASURES

Do not judge progress only by whether the child appears “less autistic.”

Track meaningful function.

Every 2–4 weeks measure:

☐ Sleep onset
☐ Night waking
☐ Bowel frequency
☐ Abdominal pain
☐ Food variety
☐ Hydration
☐ Sensory tolerance
☐ Recovery time after overwhelm
☐ Communication
☐ Self-injury
☐ Aggression
☐ School participation
☐ Ability to tolerate transitions
☐ Toileting
☐ Family stress
☐ Child's individualized goals

Progress might mean:

“Episodes still occur, but recovery decreased from 60 minutes to 15 minutes.”

That is meaningful improvement.

XXIV. PRACTITIONER DOCUMENTATION LANGUAGE

Recommended wording

NLS/Biofeedback Observation

> Biofeedback screening was used as a non-diagnostic wellness assessment. Findings represent patterns generated by the biofeedback system and are being used to identify areas for further questioning and supportive wellness planning. They are not interpreted as confirmation of disease, infection, allergy, genetic abnormality, or organ pathology.

Referral wording

> The current wellness assessment raised questions that fall outside the scope of holistic wellness education. The family is encouraged to discuss these findings and symptoms with the child's pediatric healthcare provider and/or an appropriate specialist.

Integrative wording

> The Holistic P.E.N. Way™ plan is intended to complement, not replace, appropriate pediatric, developmental, neurologic, genetic, occupational therapy, mental-health, nutritional, or other professional evaluation.

XXV. THE PRACTITIONER'S ROOT-CAUSE MAP

When a child comes in with autism, sensory difficulties, behavioral dysregulation, developmental differences, ADHD-like symptoms, trauma symptoms, or other neurodivergent traits, investigate these pathways:

1. Developmental

↓

2. Genetic / Chromosomal

↓

3. Neurologic / CNS

↓

4. Prenatal / Perinatal

↓

5. Sensory Processing

↓

6. Autonomic / Regulation

↓

7. Trauma / Somatic

↓

8. Sleep

↓

9. Nutrition / B Vitamins / Minerals

↓

10. Gastrointestinal

↓

11. Allergy / Food / Environmental

↓

12. Endocrine / Metabolic

↓

13. Infectious / Parasitic when clinically indicated

↓

14. Environmental Exposure when clinically indicated

↓

15. Family, School & Environmental Stress

↓

Then ask:

Which of these factors are actually present in THIS child?

Not:

Which theory can explain every neurodivergent child?

Holistic P.E.N. Way™ Core Principle

REVEAL WHAT'S HIDDEN • RESTORE WHAT'S BROKEN • RENEW THE WHOLE PERSON

For neurodivergent children, “reveal what is hidden” should mean identifying overlooked pain, nutritional insufficiency, sensory overload, sleep disturbance, trauma, developmental needs, neurologic concerns, gastrointestinal problems, genuine infection, genetic conditions, environmental stressors, and communication barriers.

The goal is not to erase the child's identity.

The goal is to **remove unnecessary physiological and emotional burdens and give that child's brain and body the safest possible environment in which to develop, regulate, communicate, learn, and thrive.**